Smart Diet Management through Food Image and Cooking Recipe Analysis

Zeynep Hakguder
School of Computing
University of Nebraska - Lincoln
Lincoln, NE, United States
zeynep.hakguder@huskers.unl.edu

Yinchao He
Jacob School of Engineering
University of California - San Diego
La Jolla, CA, United States
yih039@ucsd.edu

Weiwen Chai
Department of Nutrition and Health Sciences
University of Nebraska - Lincoln
Lincoln, NE, United States
wchai2@unl.edu

Juan Cui*
School of Computing
University of Nebraska - Lincoln
Lincoln, NE, United States
jcui@unl.edu

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***Abstract*—Food monitoring has become an indispensable practice for personal health management in increasingly growing populations. To facilitate this process, advanced image processing and AI technologies have been applied to empower automated recognition of food item and nutrient using food images taken by smart mobile devices. However, existing tools suffered from unsatisfactory precision and compromised convenience, which has hindered broad application of food logging tools. In this study we explore new solutions to image-based food recognition for improved performance, with a particular focus on domestic cooking applications. Particularly, we leverage advanced machine learning and nature language processing techniques, in conjunction with comprehensive food nutrient profiles in the knowledge base and the contextual ingredient information parsed from publicly available recipes, in developing a new food recognition system. Our optimized models were proved to be effective in ingredient recognition based on food images and is under the integration into an Android app named FoodInsight, available at https://github.com/zhuxinyishcn/FoodInSight.**

***Index Terms*—food logging; image recognition; food recommendation; recipe analysis; domestic cooking; transfer deep learning; natural language processing**

I. Introduction

High obesity rates and the associated risk of other serious health conditions such as diabetes, metabolic syndrome, and cancer can be mitigated by effective food monitoring and diet management. While food journaling has been a normal practice in the healthcare domain for decades and has inspired inventions of numerous facilitating software tools, it often requires large effort on the user's part and significant manual effort required by the existing systems can deter the user from continual usage. To this end, automated food recognition using dish pictures taken by mobile devices can simplify the logging process and offer ease of use. It is expected that new innovative health assistant systems can analyze food items and their macronutients based on the food images and ideally can provides recommendations of healthier and compatible ingredients to user's meal.

Recent work on imaged-based food recognition mainly applied deep neural networks (DNNs) for classification [16], [19], [24]. From a diet monitoring perspective, two major challenges are present in the current research towards generating a generalized classification model. First, performance of machine learning (ML) models including DNNs are limited by the quality of the datasets they are trained on. The training dataset should contain a large number of images closely related to the task. To this end, a few large food datasets were created and ready for ML tasks, such as Food101 [1], UEC-FOOD100 [22], UEC-FOOD256 [17], which contain dish images of American and a few international cuisines. However, such food images are mostly taken in a well-controlled manner, specific to certain types of cuisine, and with food items easier to separate. Although the image-classification performance can be promising, it is often difficult to assess food from a dish image with mixed items, especially when items are overlaid. Second, food portion estimation directly from images remains challenging. Current approaches tackle this problem by attempts using predefined 3D food models [8], predicted depth maps of eating scenes [23], or energy distribution maps [24]. Note these solutions estimate the content of the dish image either as a whole or through simplified geometric models that need to be expanded for practical use. Additionally, the accuracy and feasibility of some solutions are hindered by the dependence on additional devices that are inconvenient to use or costly. When the task is performed in an uncontrolled setting in actual home cooking where the food items are not limited and standard to a specific cuisine and some items might be occluded in the images, all these challenges get compounded.

There are other important factors to consider in order to make an automated diet monitoring application practically useful. For instance, while extremely valuable, caloric content of a meal is not comprehensive enough to provide insightful

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feedback to a user on their nutrition goals. Users from different background might pay special attention to the balance of their macronutrient intake in carbohydrate, fat, protein, and etc. A detailed profile of different nutrient information is clearly important to guide the user toward healthier diet choice, in a way that better serves their individualized needs.

To address aforementioned challenges and improve the food recognition accuracy particularly in domestic cooking, we aimed to explore new solutions in this study by leveraging advanced ML techniques, comprehensive nutrient profiles of food items in the knowledge base, and contextual, qualitative, and quantitative information of ingredients parsed from known recipes. Specifically, the core represents a fundamental model for imaged-based food classification. Given a high dimensional image, the aim of classification is to assign the most likely label/class name to each food item identified on the image. Features used to predict the image class can either be engineered as how it was done in traditional computer vision approaches or be learned by automated algorithms such as convolutional neural networks (CNNs). The new model has to collectively classify multiple raw food ingredients prepared for cooking a meal based on a given image. Different from existing multi-class classification systems, an unique design of our model is to consider the dependency within the labels, an important property of so-defined structured prediction problems. In this case, classification of ingredients will be reevaluated based on the probability that all labeled ingredients are present in the same meal together. Such probabilities can be estimated by examining known recipes collected from public sources. The benefit of this design is that it takes advantage of ingredients that the model has confidence in prediction and the co-occurrence information of ingredients in known recipes to ameliorate predictions for which the classifier is not highly confident. To this end, we will find the set of labels that maximizes the conditional probability using an approach based on an emerging technique in natural language processing (NLP) called conditional random field (CRF). The developed models are integrated as part of an in-house smartphone-based diet management, named FoodInsight. While the app prototype is still under a feasibility user test, we have evaluated the performance of the presented models through different experiments in this study.

The rest of this paper is organized as follows. In Section II, we present an outline of the related work. In section III, after an overview of the entire approach, detailed description of the methodology and datasets are provided. Section IV provides the results and discussion, which is followed by the conclusion and future perspective.

II. Related Work

Early attempts at food image recognition focused on feature engineering. One of the first such attempts was to train Support Vector Machine (SVM)-based classifiers using images of the newly-compiled Pittsburgh fast-food image dataset (PFID) and features obtained by the color histogram and Scale-Invariant Feature Transform (SIFT) [5]. Although the performance is overall limited, further accuracy improvement (19% to 28%) was soon obtained using multiple kernel learning algorithms with features such as color, texture and SIFT information [15]. In addition to having manually created features, those earlier studies have used images restricted to having a single food item on the dish or multiple items that have been manually segmented. Since then, several new datasets and ML models have been developed. For examples, UEC-FOOD100 [22] dataset contains images of 100 Japanese dishes, UEC-FOOD256 dataset contains [17] images of 256 international dishes, and Food-101 [1] dataset contains images of 101 dishes. With a random forest classifier, 50.7% accuracy was achieved on the Food-101 dataset [1] while more recently, CNNs were primarily applied in classification of food images from these databases [6], [10], [29], [33], [34]. Overall, the image-based food classification remains challenging as the performances of current models are way far from satisfactory.

When classifying images taken from a mobile device, classification performance can be improved using contextual information such as user-stated preference or similar meals taken in the past. Other source of contextual information includes co-occurrence of food items in the same meals. In [12], the co-occurrence information was modeled by a fully connected undirected graph and the classification achieved a 44% accuracy over 56 categories of images collected in natural eating conditions. The co-occurrence model has to be updated every time when new images are added to the collection. More recently, Myers et al. predict both the food items in a given meal and their nutritional content in and out of the restaurant setting using a multi-label CNN classifier [23]. In the restaurant setting, the predicted food items are used to look up restaurant menus to retrieve the nutrition content. Out of the restaurants, estimation of the nutrient content is more difficult due to the challenges in volume estimation. To this end, the authors crowd sourced manual segmentation of images and compiled a segmented dataset of 201 food items to be used for volume estimation in conjunction with depth estimation by another CNN model. Their segmentation system wasn't successful on real food data but the depth prediction architecture was able to reach close-to-groundtruth depth values. In addition to the aforementioned classification-based systems, Sun et al. used the YOLO [25] architecture to detect what food items are in the image and report their positions. By looking up a nutrition database, the system can further obtain the per serving nutrient content of each item [30]. Note the image dataset is restricted to mostly Japanese dishes. Another recent development worth-mentioning is the food classifier developed by Facebook [26]. For a given prepared food image, the food classifier lists ingredients that were used to prepare the dish in the image. Since both ingredient images and recipes associated with the images were used during the model training process, the classifier learned to associate prepared dish images with individual ingredients.

Another line of work is to generate or retrieve recipes based on the identified food images. One study trains a decoder with a deep learning architecture on both extracted visual

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representations from CNNs and the embeddings calculated for the predicted ingredients from the image, to generate recipes with step-by-step instructions [26]. Similarly, Chen et al. propose a method for recipe retrieval by first predicting the ingredient types and the cutting and cooking attributes [3]. Neural networks with attention mechanism have also been employed lately on this topic. For example, word- and sentence-level attentions on the recipe representations were combined with attentions from the corresponding image parts to identify image-recipe pair that shares closer embedddings [4]. Additionally, more complex architectures, such as double-triplet scheme have been presented in [2], [27] for recipe retrieval problem to learn joint embeddings of the image and recipe pairs. Their work was evaluated on Recipe1M+ dataset which has been compiled for learning cross-modal embeddings for food images and recipes [21]. Other recent examples include a study that uses a CNN architecture on images together with long short term memory (LSTM) and bidirectional LSTSMs on recipes to generate cooking instructions for Indian dishes [18] and another one presenting a multi-modal approach to incorporate cuisine style, course type, and ingredient types or flavors into the model to enable the flavor analysis and recipe recommendation [13].

III. METHODOLOGY

A. Overview

This section overviews the workflow of our approach with more details explained in each subsections. As illustrated in Fig. 1, our system is designed for image recognition in a domestic cooking. In this setting, image of raw ingredients prepared for cooking will be provided. During mobile usage, a 3×3 grid is overlaid on the camera to guide the placement of ingredients in the picture to facilitate ingredient classification. The image classification was performed in two stages.

First, a DNN model was used to obtain probabilities for each of the different food classes. In particular, a supervised classification model was trained based on 130 families of food items and then used to assign labels to each food images. Here, each item image x_i is represented by a $256 \times 256 \times 3$ vector while a label y_i is represented by one-hot encoding which is a binary vector of length 130 with every entry being 0 except at the position corresponding to the true class of the item. This problem is solved by supervised training of a classifier based on N data points containing both images and their labels, $\mathcal{D} = \{\mathbf{X}_i, \mathbf{y}_i\}_{i=1}^{N}$. After the model was optimized, we used it to label each food item on an image along with the inferred probabilities.

Second, these label probabilities were combined with cooccurrence probabilities of ingredients in recipes to obtain the optimal classification of the multiple item images as a whole. Particularly, CRF was applied to model the conditional probability and look for the set of labels $\mathbf{y}^* = \underset{\mathbf{y}}{\operatorname{argmax}}\, p(\mathbf{y} \mid \mathbf{X})$ that maximizes the conditional probability. Once the ingredients are identified, either from pictures of raw ingredients or of

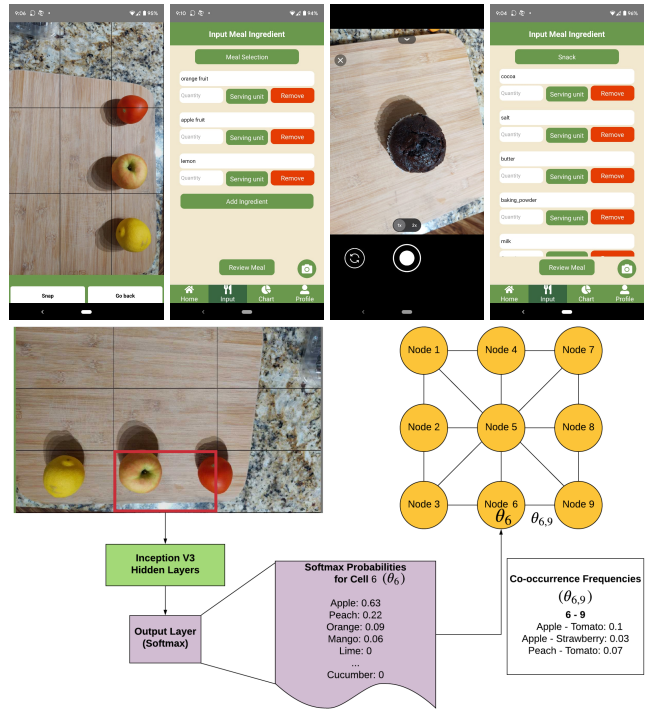

Fig. 1: (top) Predicting based on raw ingredients using the grid guide vs prediction based on prepared meals; (bottom) illustration of the prediction workflow. Each cell in the grid may contain an image of an ingredient. These images are fed into a neural network for classification individually. The softmax probabilities output by the neural network are used as node potentials in the CRF. Edge potentials are calculated from the recipe corpus and represent co-occurrence frequencies of pairs of ingredients together in the corpus.

whole prepared plates, we will search the recipe database to find most similar recipes as recommendations.

To extract recipe-related information, we have applied different NLP methods to handle various issues. The first task is a named entity recognition problem, which is identification of phrases made up of one or more words that refer to an entity along with their entity type in some text. These types can be organization, name, country, etc. In our case, we want to identify parts of a recipe ingredient item, where an item is composed of quantity, unit, ingredient name, and comments. The second NLP task is to cluster similar ingredient names together. This is done so as to combine together ingredient names that are considered the same for our purpose. For instance, we do not wish to distinguish between "cinnamon" and "cinnamon stick", for our purposes it suffices to have both under a cluster named "cinnamon". The last NLP task is to find a set of similar recipes to a query recipe. At this point, each recipe in our dataset is transformed into a vector where each entry corresponds to a ingredient cluster obtained in task two. We approach this problem by using cosine similarity between the query recipe vector and the recipe vectors in our database.

Those major procedures are detailed in each of the following

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subsections.

B. A deep learning (DL) model for ingredient image classification

1) Ingredient image dataset: In order to train a ingredient recognition mode, we first collected images of ingredients that are commonly used in domestic cooking, which resulted a dataset containing 86,265 images on 130 classes of ingredients including vegetables, fruits, meat, cheese, egg, and etc. Each class has at least 240 and on average 663 images. The correlated images are collected from Google Images using ImageAssistant Batch Image Downloader, a free image extractor. Each image in the dataset is at least 100x100 pixels. This dataset was randomly divided into 80% *training*, 10% *testing* and 10% *validation* sets per class.

2) Training the classification model via transfer learning: Next, we transfer learned a classifier from InceptionV3 CNN architecture [31] that is known with high classification performance and less costly training compared to other high performance CNNs. This architecture incurs less computational cost without a sacrifice on the classification accuracy by factorizing convolutions. They used a series of 3×3 convolutional filters instead of larger filter sizes and asymmetrically factorized of $n \times n$ filters into $1 \times n$ and $n \times 1$ filters. We started training the neural network with weights set to ImageNet [7] weights. We replaced the last classification layer of InceptionV3 with 130 nodes each corresponding to one of the ingredient classes.

We used a learning rate of $\alpha = 0.01$ with RMSProp optimization algorithm and set the number of training epochs to 1000. Early stopping applied as regularization with a tolerance parameter set to $t = 10$. Tolerance parameter determines how many epochs the network continues training without any improvement on validation accuracy.

Further, we have explored four other deep learning architectures on our dataset: ResNet50 [11], EfficientNet [32], DenseNet [14], and VGG16 [28] as a comparison.These networks were transfer learned from weights learned on ImageNet dataset.

C. Recipe processing using NLP

1) Recipe dataset: Over 100GB of recipe data is crawled using Python3 and Beautiful Soup from Allrecipes.com, which includes a total of 10,421 recipes of appetizers, desserts, main dishes, and soups from 21 types of cuisine from American, Argentinian, Australian and NewZealand, Brazilian, Canadian, Chilean, Chineses, Colombian, French, German, Greek, Indian, Israeli, Italian, Korean, Mexican, Peruvian, Spanish, Swiss, Thai, and UK. For each recipe, it consists of five components including recipe ID, recipe name, ingredients, serving information, and nutrients.

2) Ingredient name recognition, clustering, and cooccurrence estimation: The first NLP task is to identify parts of a recipe ingredient item, where an item is composed of quantity, unit, ingredient name, and comments. An example ingredient item looks like below with the words in bold to differentiate from entity categories.

1 (quantity) - **cup** (unit) - **of** (comment) **shredded** (comment) - **cheese** (name)

We first used a linear chain conditional random field to address the task of assigning a label from {"quantity", "unit", "name", "comment"}. We used 17 unigram templates and 1 bigram template to generate features on our data following [9]. Unigram templates produce feature functions for the connections between output at t and input in the graph, while bigram templates produce feature functions between outputs at t and $t-1$. In Fig. 2, unigram features are shown in red, and bigram features are shown in blue.

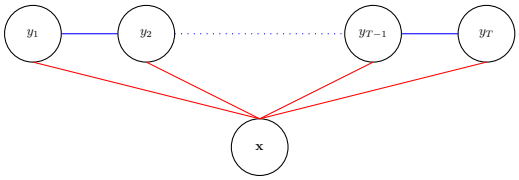

Fig. 2: A linear chain conditional random field with input $\mathbf{x}$ and output $\mathbf{y}$

The feature functions are defined using properties of the input. We used five properties, which are capitalization (C), if the input word is parenthesized or not (P), cap (CAP) on the limit of the ingredient list item (one of $\{4, 8, 12, 16, 20, 20+\}$), position (POS) of the word on the ingredient list item, and the word (W) itself. We list the unigram templates used in Table I.

TABLE I: The template list for unigram features. The first section lists the features defined on single property of the input. Second section lists the features depend on property pairs of the input.

Position	Property
Two positions back	W,P
One position back	W,P
Current position	W, POS, CAP,C, P
One position after	W,P
Two positions after	W,P
Current position	W&POS,W&CAP, POS&CAP,W&C,W&P

We give an example input ingredient item from our dataset with its associated properties and BIO tag in Table II. In our application, we want to find spans of text that are of interest and label those spans. BIO tagging [**?**] is a standard approach that transforms the span labeling problem to a word labeling problem. Its labels capture both the position information of the word in an entity and the type of the entity

TABLE II: An example input with properties

Input	POS	CAP	C	P	BIO tag
1	1	8	0	0	B-QTY
Cup	2	8	1	0	B-UNIT
Mozzarella	3	8	1	0	B-NAME
,	4	8	0	0	OTHER
shredded	5	8	0	0	B-COMMENT

Let current word be "Cup", some feature functions on the above examples are:

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- Current word is "garlic" and output is B-QTY
- Current word is "shredded" and output is B-QTY
- Current word is "tomato" and capitalized and output is B-UNIT

All above feature functions evaluate to false. Feature functions below evaluate to true:

- Current word is "cup" and output is B-UNIT
- Current word is capitalized and output is B-UNIT
- Next word is Mozzarella and output is B-UNIT

We trained our CRF model on New York Times cooking dataset released in 2015 [1] using 20,000 ingredient items and tested on 2,000 ingredient items.

Next, we used string matching to group similar ingredients together after performing named entity recognition on our recipes collected from "AllRecipes.com". We started by removing plurality suffixes. Then, we divided the multi word ingredient names into their word constituents. We assumed the first and last words in an ingredient name would be most informative about the ingredient such as in "vine tomato" or "fig jam". We started our groups with single word ingredient names in our dataset and considered each single word ingredient name as the keyword of a group of ingredients. For ingredients with multiple names, if any word in the ingredient name agreed with the keyword of a cluster we placed the ingredient under that cluster. While doing that we prioritized the first and the last word in the ingredient name following our assumption.

Once all ingredient names have been extracted from all recipes, we calculate the co-occurrence frequencies between ingredient $iand$j and the probability is estimated as the ratio between counts of receipts that have both $iand$j and count of recipes that have either iorj.

D. Optimized classification via CRF

A CRF graph is built based on each identified ingredient. We let the node potentials come from our neural network classifier with InceptionV3 architecture and edge potentials are the normalized co-occurrence probabilities of ingredients calculated from the recipes. This graph is fully connected and to reduce the computational cost we limit the candidate ingredients in each position to five for which we pick the candidates with top softmax scores from the predictions of the neural network.

Optimization was performed to determine the optimal set of ingredients based on both predicted probabilities and the co-occurrence probabilities in recipes. We provides the description on recipe processing to prepare such required information.

As an independent validation, we downloaded the Food.com dataset which contains both images and recipes (raw and processed) [20]. In total, the dataset contains more than 180,000 recipes and 13,627 ingredients. We removed rare ingredients which were listed in fewer than 100 recipes. We also removed entries that are not ingredients, e.g. tomato soup, aluminum foil. Further, we mapped each ingredient to a super-ingredient, e.g. shredded beef was mapped to beef. This images from this dataset will be subject to the food recognition system and the identified recipe will be compared to the prediction results to examine if the prediction is desired at the recipe level or nutrient level.

[1] Available at https://github.com/nytimes/ingredient-phrase-tagger/blob/master/nyt-ingredients-snapshot-2015.csv

E. Recipe recommendation

To explore if alternative cooking recommendation can be recommended based on the set of identified ingredients, we designed the following strategy to search for recipes that are most similar to the predicted one (the query). One important task is to differentiate recipes that use an ingredient in the query set as the dominant ingredient from those that use them as supporting ingredients. For example, both a quiche and a roux contain flour but the role and amount of the flour is different in these two foods.

First, we represented each ingredient cluster c_i in the recipe by its weight $w(c_i)$ and mark the more significant ingredients, the most abundant ones. In our raw recipe dataset, both volume and weight units as well as non-standard units such as 'a sprig' were used. We converted all units to weight units (grams) for each type of ingredient (Supplementary methods) and then define the weight of an ingredient cluster $w(c_i)$ in a recipe r_j as the total weight of all ingredients in r_j that are in c_i.

$$w(c_i) = \sum_{i_j \in r_i} w(i_j),$$

where $w(i_j)$ is the weight of ingredient i_j in recipe r_j. To bring all recipe candidates to the same scale, we normalized the weights of each cluster by dividing by the total mass of the recipe r_j;

$$w_s(c_i) = \frac{w(c_i)}{\sum_{c_k \in r_j}}$$

Further, we considered having common rare ingredients is more indicative of similarity between two recipes. We used the negative log probability $e_i = -\log p(c_i)$ of an ingredient cluster as a rareness metric which increases as the frequency of the cluster decreases and hence serves our purpose well. Probability of a cluster $p(c_i)$ is equal to the proportion of recipes that contain at least one ingredient from cluster c_i.

$$p(c_i) = \frac{|\{r_j | r_j \text{ contains an ingredient in } c_i\}|}{|\mathcal{R}|}$$

The final similarity value s_i for cluster c_i is defined as

$$s_i = w_s(c_i) * e_i.$$

Finally, similarity of recipe r_i to a given predicted set of ingredients I is the sum of the similarity scores for all ingredient clusters in I, $s_{r_i} = \sum_{c_j \in r_i} s_j$.

F. Ingredient volume and nutrient estimation

After identifying all food items from an image, it is important to assess the nutrient content, e.g., amount of carbohydrates, sugar, proteins, lipids, and calorie, which will require weight estimation. In a previous study (Bruno's paper),

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we have applied an approach each classified food item will have its area (by the top picture) and height (based on side images) estimated using user's fingertip as a reference object. The fingertip and the food items need to be positioned in specific areas when taking the pictures. After segmentation of the top image, we have the total pixels that form the finger area and the food items, respectively, ready for the estimation of the size of each food area. Based on the side pictures. one can estimate the food height after segmentation. The final step is to multiply the area by height, which results in food volume. Our system contains a database of food items with correlated volume/weight and nutrient information, which was collected based on a nutritional facts table from the USDA Food Composition Database [29] and can be directly used for the calculation of nutrient content based on the estimated volume and weight.

As an alternative strategy that bypasses volume estimation, we can estimate nutrition content in the food images using information from recipes. Our nutrition estimation approach first finds k most similar candidate recipes, following the procedure explained in the above section and lists the ingredients in these recipes along with their amounts. At this point, the user gets a chance to choose the most similar recipe in terms of ingredients and modify the ingredients and their amounts in the recipe. Lastly, the nutrition content is looked up in our database per 100 gram of the listed ingredients and scaled according to the amount listed in the final ingredient list. The unit conversion is provided in the supplementary document.

IV. Results

A. Food classification with InceptionV3 outperforms other architectures

We stopped training of InceptionV3 model when it stopped improving after achieving 87.47% validation accuracy with 87.63% testing accuracy. Several classes were classified with perfect score of 100%, e.g., ginger, lobster, crawfish, spaghetti squash. Monterey-Jack cheese was the class with lowest accuracy of 25% (Fig. 3). We further investigated the breakdown of

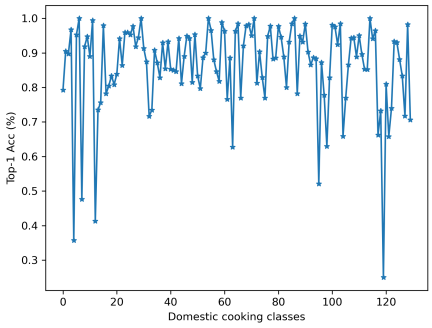


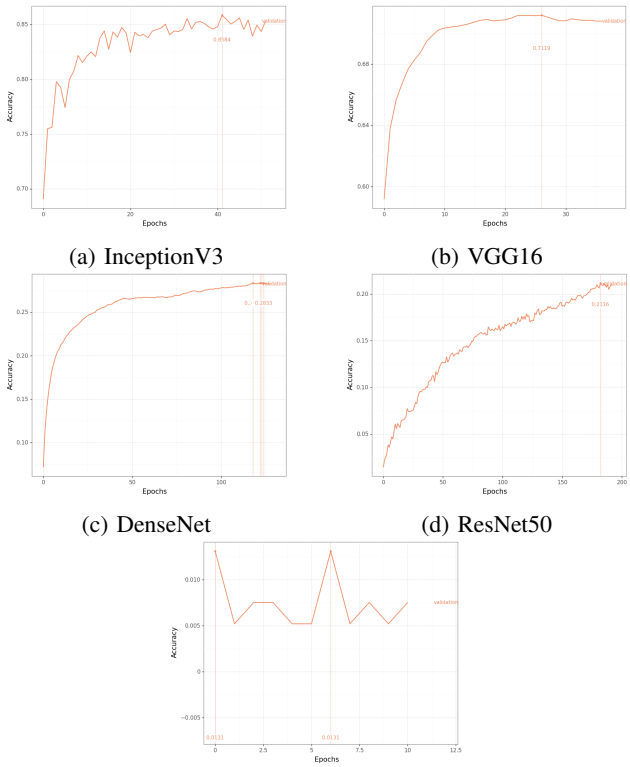
Fig. 3: Top-1 accuracy of Inception model on domestic cooking dataset. Evaluation was done on the validation set.

misclassified classes. The lowest performed class Monterey-Jack was classified solely as dairy products: butter, gouda, Swiss, pepper Jack and Colby Jack. Second worst class is chuck roast beef classified as other cuts of steak: T-bone, ribeye, tenderloin, eye of round steak and as pork loin chop.

While misclassifications listed above do not cause serious problems for nutrient estimation, there are other classes that are mixed based on their similar visual characteristics, however nutritionally they are different. 13% of the parsley was classified as endive and 13% as kale. There are other cases for this type of misclassification in the validation dataset such as bananas classified as yellow squash, apple as shallot, white onion as melon, cucumber as zucchini. There is room for improvement in cases like these.

To compare with other deep learning architectures, we make sure all networks were transfer learned from weights learned on ImageNet dataset.

Fig. 4 shows validation accuracies over epochs. All network architectures except EfficientNet improve over time. However, DenseNet, ResNet50, and VGG16 plateau at significantly worse accuracies that InceptionV3. Hence, as our final model we use InceptionV3 architecture with weights frozen at the highest accuracy value.



(a) InceptionV3 (b) VGG16

(c) DenseNet (d) ResNet50

(e) EfficientNet

Fig. 4: Validation accuracies for different network architectures. The highest value of validation accuracy is marked on the plot.

We compared our classification results with food item prediction algorithms from literature in Table III. Our overall accuracy is 87.63%, which is the second best performance. It is

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important to note that the best model (with 88.28% accuracy) has much fewer classes.

TABLE III: Comparison with other DL-food classification methods.

Studies	**Models**	**Types**	**# of classes**	**Acc.**
Bossard et. al.	Random Forest	Prepared dish	101	50.7%
Ciocca et. al.	CNN	Prepared dish	73	79.0%
Hassannejad et. al.	CNN	Prepared dish	101	88.28%
Singla et. al.	CNN	Prepared dish	11	83.6%
Yanai et. al.	CNN	Prepared dish	101	78.7%
He et. al.	Nearest Neighbor	Prepared dish	56	44.0%
Our approach	CNN	Raw ingredients	130	87.63%

B. Recipe parsing via CRF

We trained our linear chain CRF with the feature functions explained in Methods section to parse the ingredient entries into "ingredient name", "unit", "quantity" and "comment". We evaluated 2,000 ingredient items on a item and word level. Out of 2,000 items, our model predicted all of the words in 1,492 items correctly. These 2,000 items contain 11,450 words. Out of these 11,450, our model predicted 10,402 correctly. Our item level accuracy is 74.64% and word level accuracy is 90.85%.

This grouping is done by a clustering process by string matching. We evaluated a few of the resulting ingredient name clusters qualitatively. This process reduced the 7,900 unique ingredient names in our dataset to 798 ingredient name clusters. We show a few representative clusters below:

- Banana: banana, banana ketchup, banana leaves, banana liqueur, banana pepper, banana pepper rings, banana peppers, banana quick bread muffin mix such the pillsbury brand, banana squash, banana blossoms, fluid banana liqueur, deveined yellow banana peppers, sweet banana peppers, yellow banana pepper, bananas, ripe bananas, very ripe bananas, firm bananas, overripe bananas, unripe bananas
- Zucchini: baby zucchini, zucchini, zucchini long, extra zucchini, flowers pumpkin zucchini blossoms, flowers zucchini blossoms, spiralized zucchini, zucchini blossoms, zucchini matchsticks, zucchini squash, zucchinis

We can see that the clustering is successful overall, however some clusters may contain ingredient names that don't belong in the clusters as well, e.g. "sweet banana peppers" in "banana" cluster, "dry rice chicken sauce mix broccoli" in "broccoli" cluster.

C. Co-occurrence information improves the ingredient recognition

Ingredient information from recipes can be used to strengthen the predictions of the Inception model in cases where the ingredients can be visually confused but their use cases are different.

We tested how the model would be effected when we incorporated ingredient co-occurrence information from recipes. Using the parser outputs for ingredient names, we calculated the co-occurrence frequencies of ingredient pairs and used these in our CRF model. We calculated the accuracy of the overall model by using n nodes, $n \in \{2, 3, \ldots, 9\}$. Our model performance declined as we increased the number of nodes in the model as shown in Table IV. Reported values are on 1,000 collages each containing n images sampled according to the co-occurrence frequencies. For each collage, we calculate the performance of the model on the collage and use these to calculate mean accuracy and standard deviation. We suspect the declining performance of our model is due to the way we process our recipes to group ingredients with same name but different representations together.

TABLE IV: Model performance with different numbers of nodes

Number of nodes	Accuracy mean	Accuracy standard deviation
2	0.79	0.29
3	0.74	0.23
4	0.76	0.20
5	0.75	0.18
6	0.71	0.17
7	0.70	0.17
8	0.68	0.15
9	0.66	0.14

D. Finding similar recipes based on food images

We evaluated our similarity metrics defined using the weight of the cluster and rareness metric. Table V shows examples of top similar recipes to a given random query ingredient set based on weight only ranking.

TABLE V: Top 2 recipes with ingredients most similar to a given ingredient query set

Query	2 most similar according to weight
pepper, onion, oil, chicken, salt, soy sauce, clove	• chicken leg quarters, olive oil, black pepper, oregano, basil, garlic powder, salt • chicken wings, water, coriander, vegetable oil, cumin, black pepper, hot chile powder, salt
pepper, chicken, carrot, onion, salt, rice, oil	• chicken broth, chicken breasts, rice, olive oil, oregano, garlic, black pepper, onion, salt • chicken broth, arborio rice, olive oil, white wine, garlic, onion flakes, black pepper, salt, parmesan cheese

CONCLUSION

In this study we have presented a new solution to automated dietary monitoring focused on domestic cooking. The major contribution is to improve food item recognition accuracy by leveraging ingredient information from known recipe, particularly in cases where two distinct ingredients are similar visually but their cooking contexts are different, on top of a reliable DL-based ingredient classification system. To the best of our knowledge, there is little research that focus on ingredient prediction in home cooking setting with much fewer classes of ingredients and comparable or worse model

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performance in terms of accuracy. Further improvement in ingredient parsing and grouping, as well as training the deep learning architecture and the conditional random field jointly can potentially lead to a better model and improved ingredient classification accuracy.

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